Database System Engineering and Distributed Backend Development

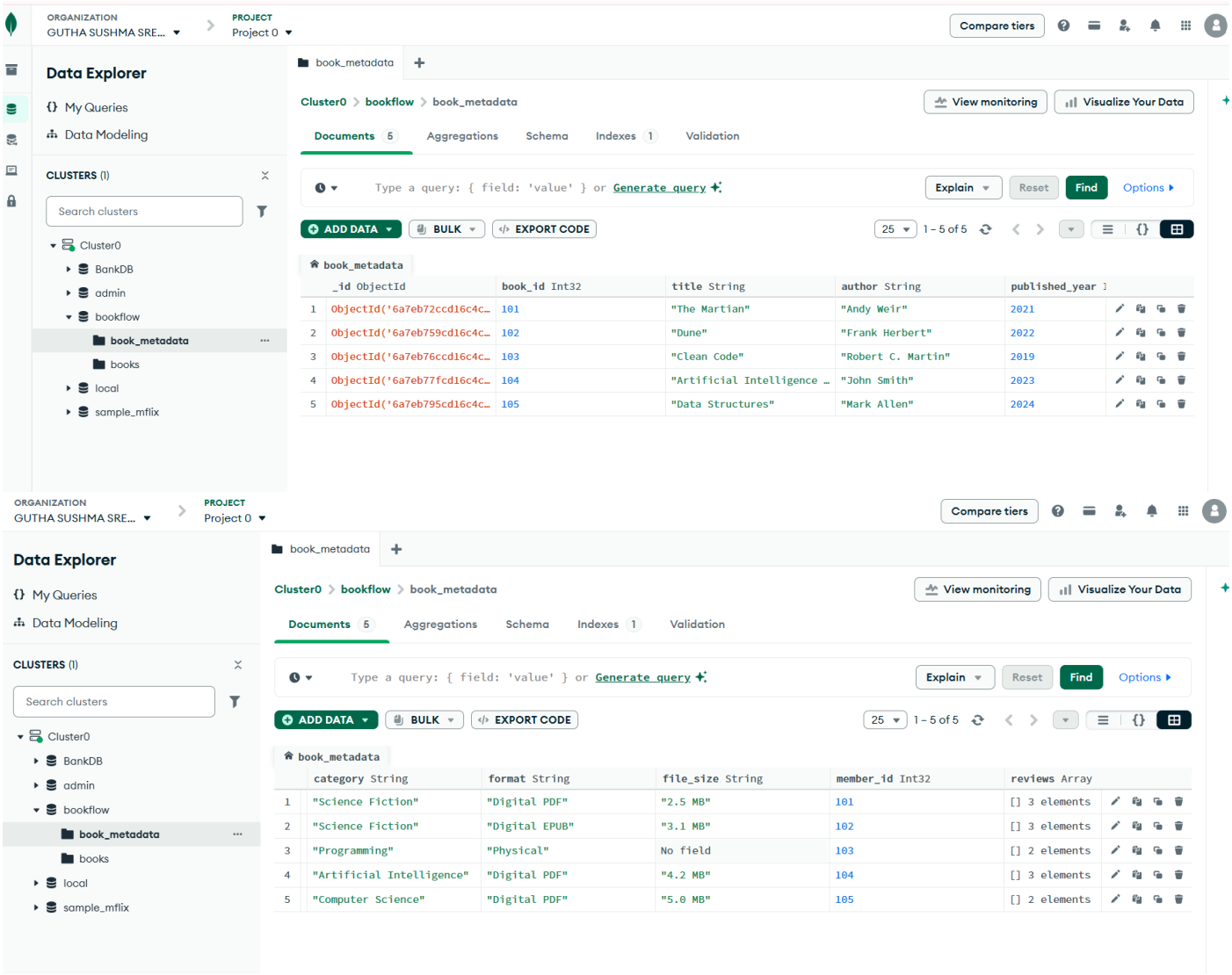
Course Code: 25CS1302E

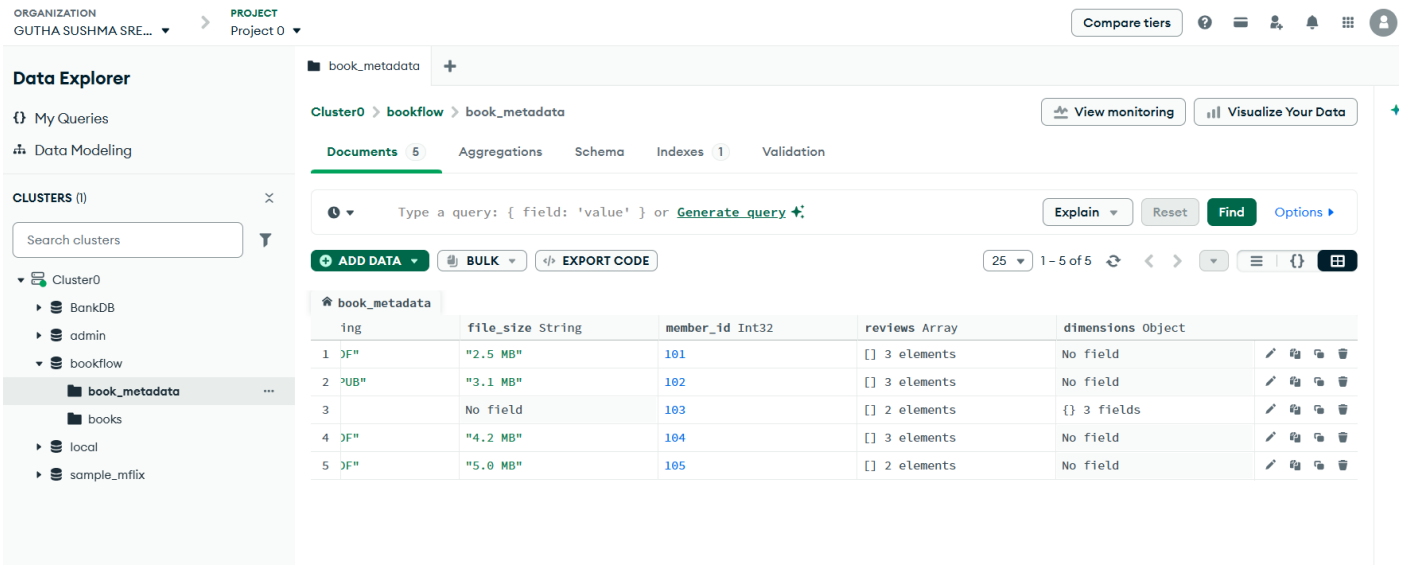
Week-3

2520030078

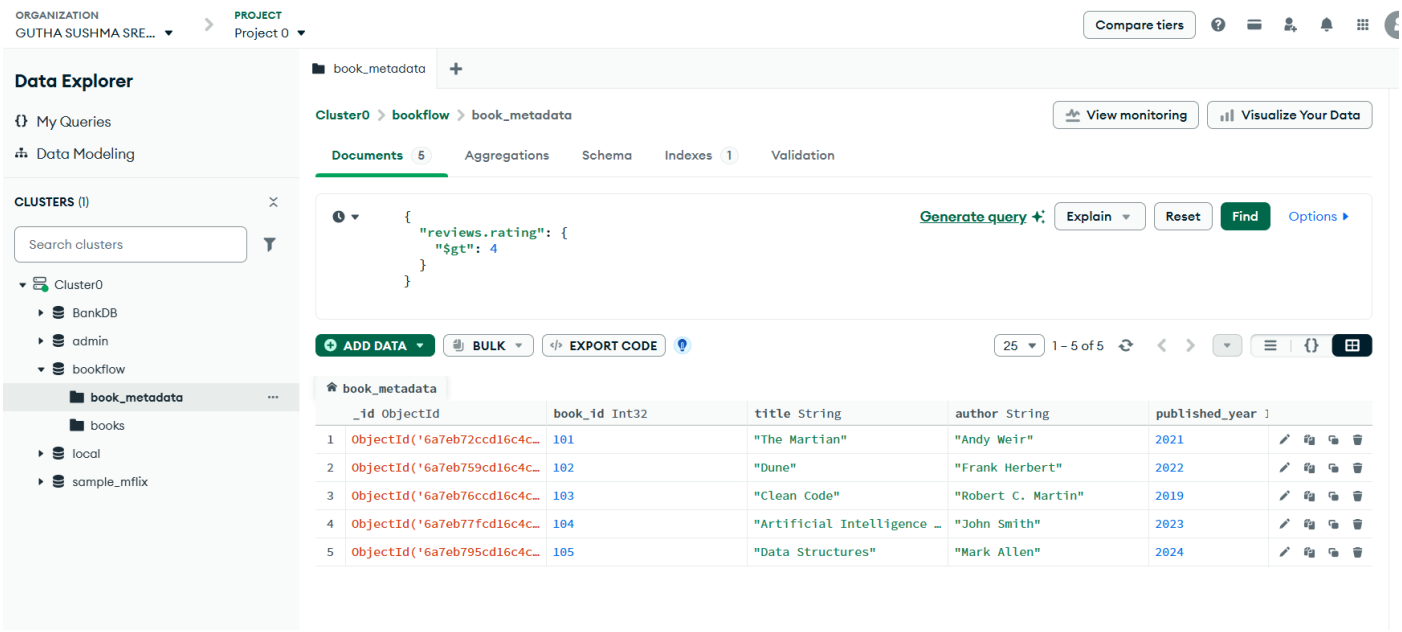
M.Apoorva

Section – 3

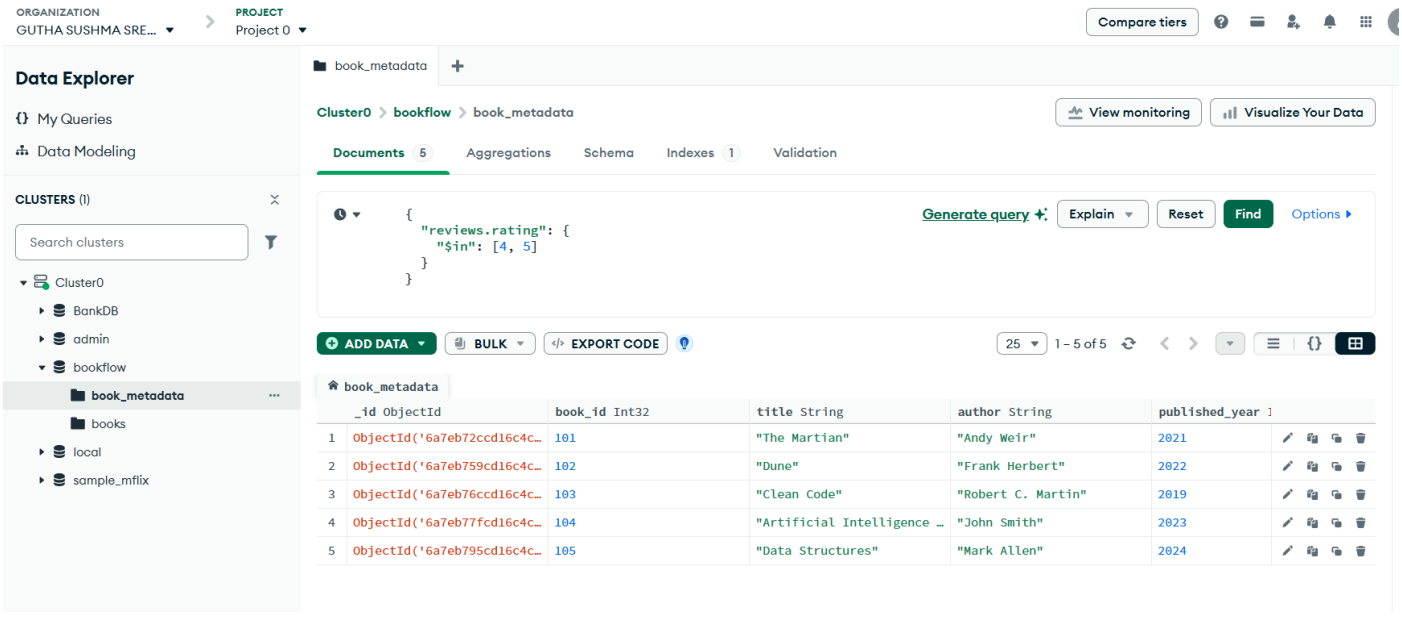




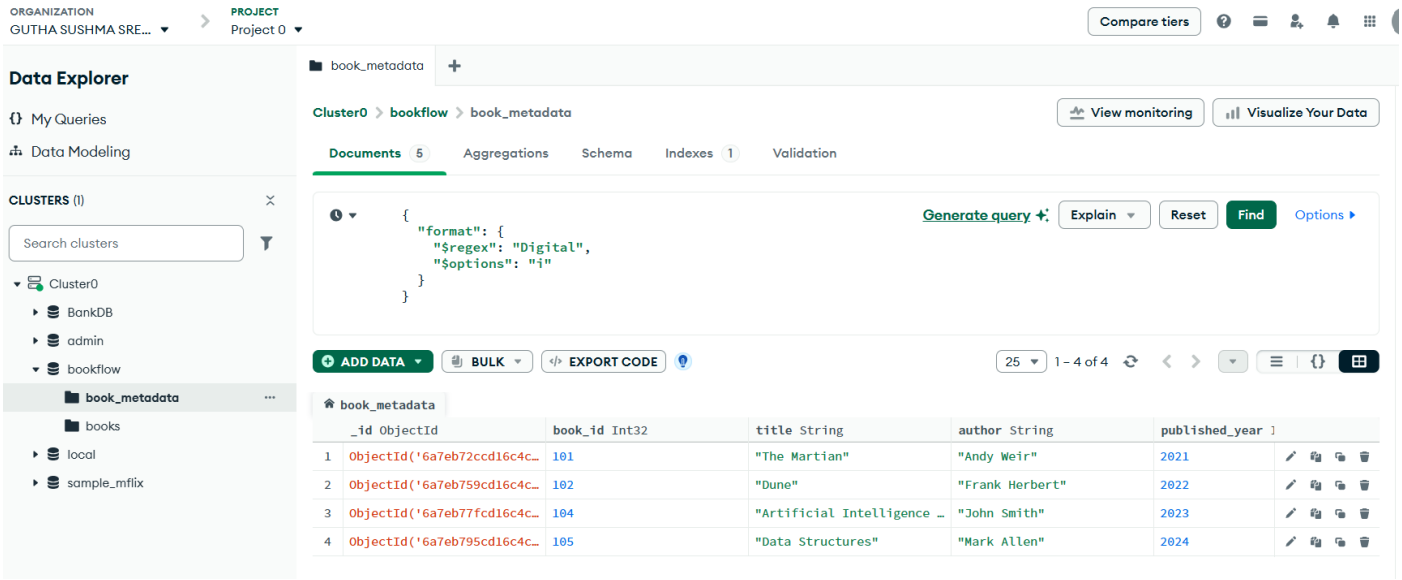
Comparison operator $gt



Using $in

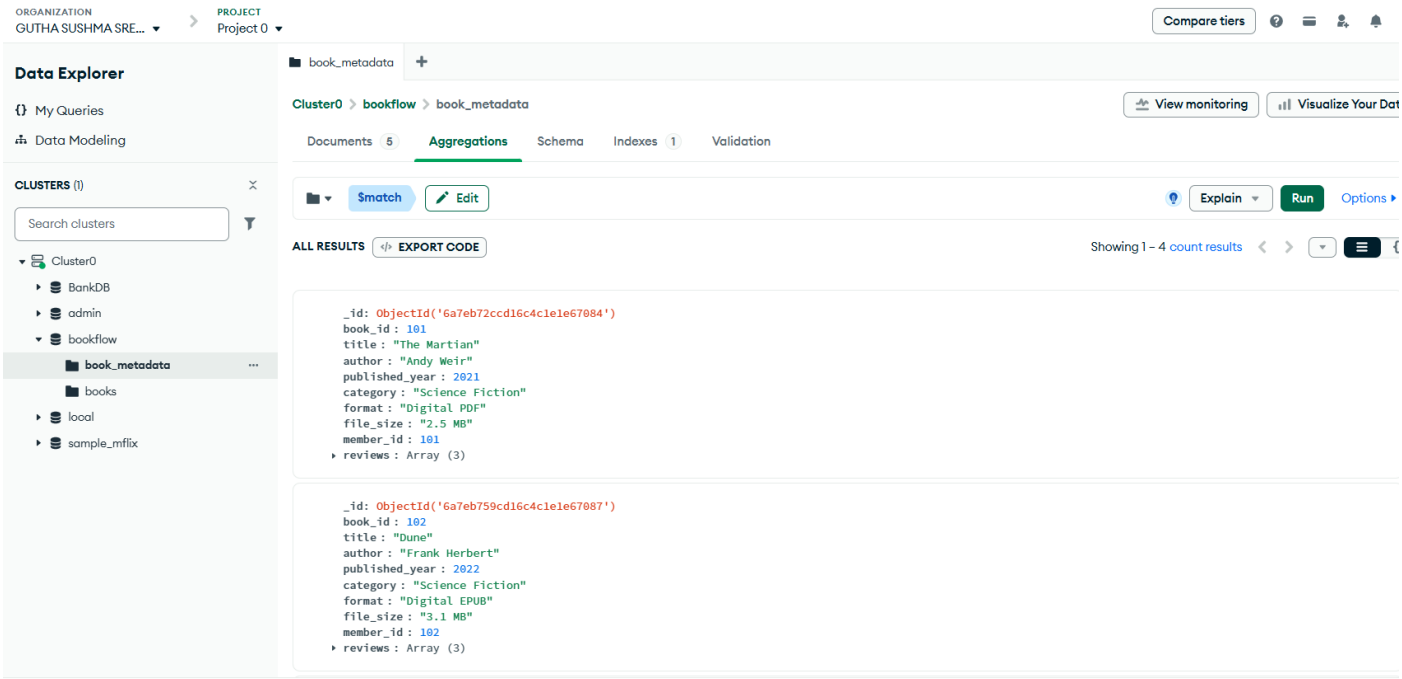


Regex query

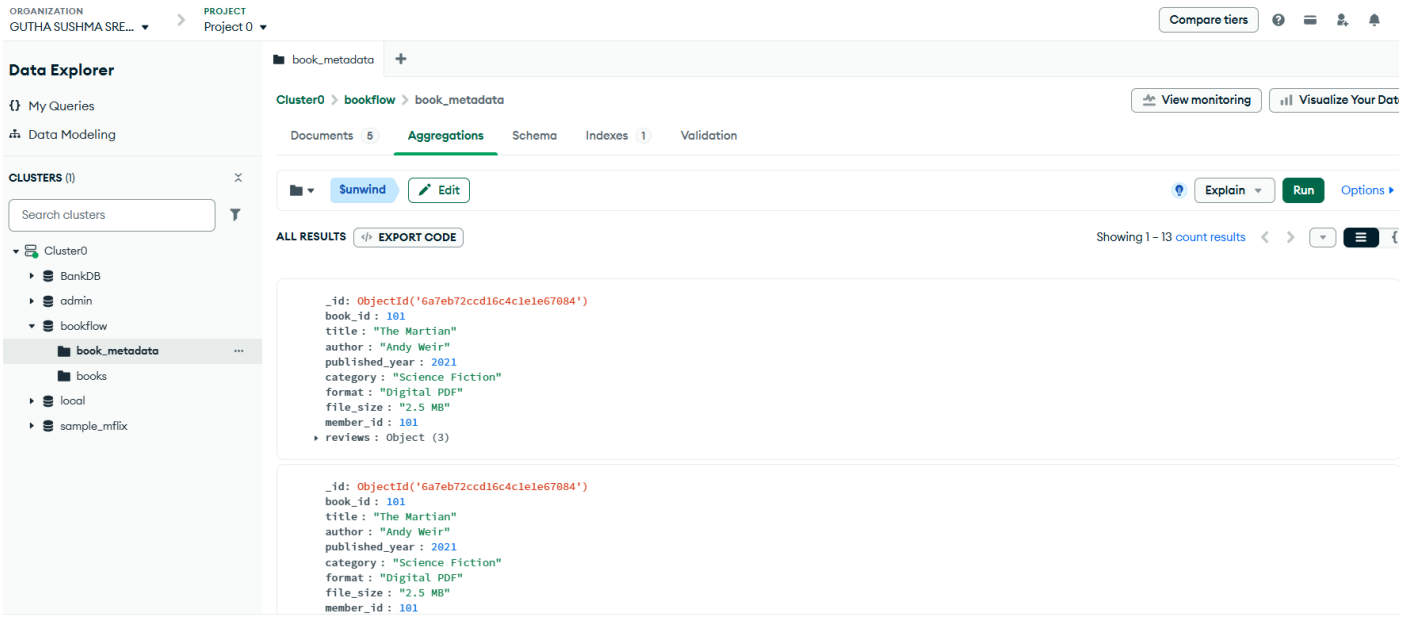


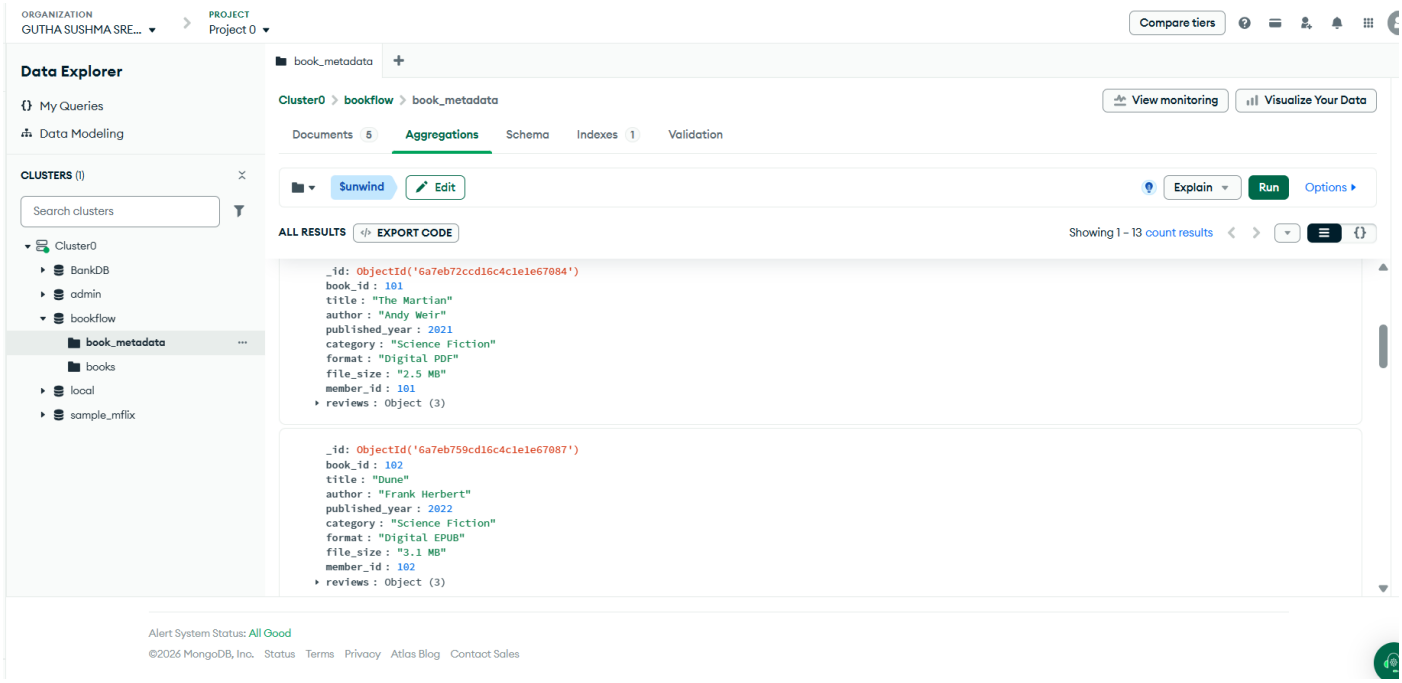
Aggregation pipeline

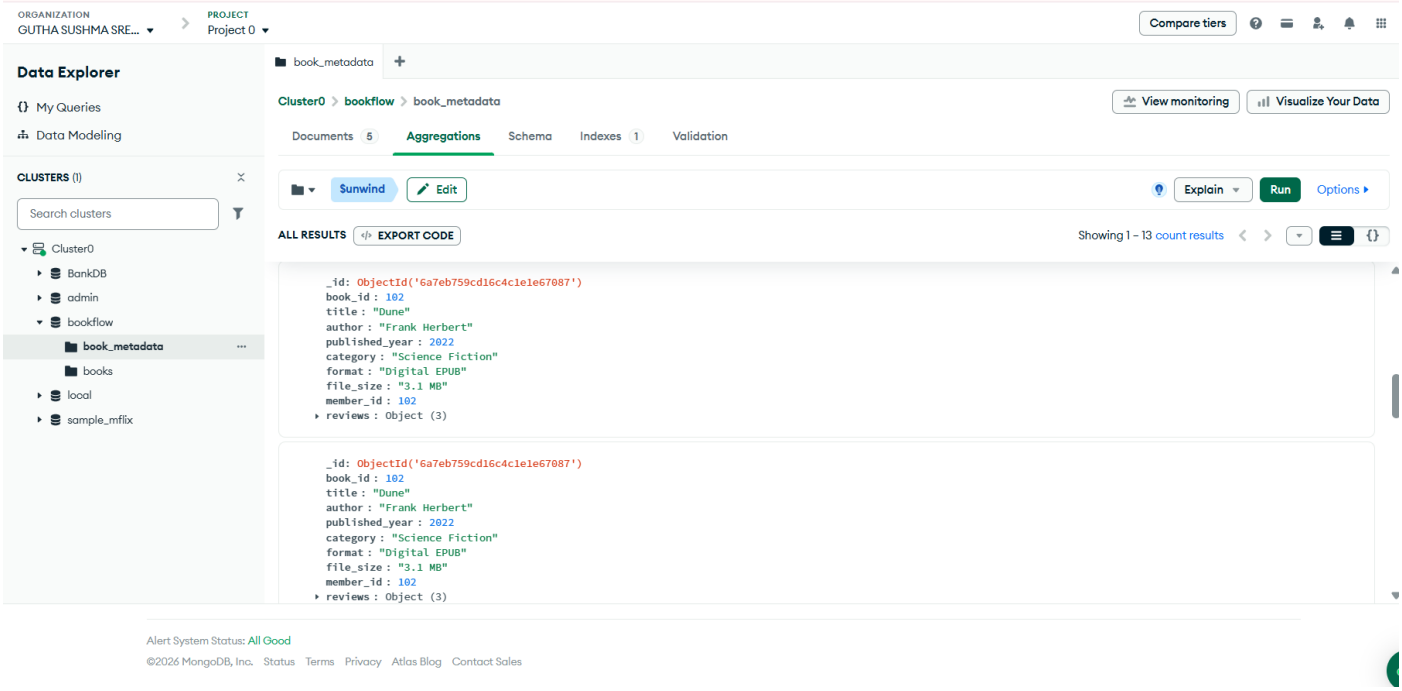
$match



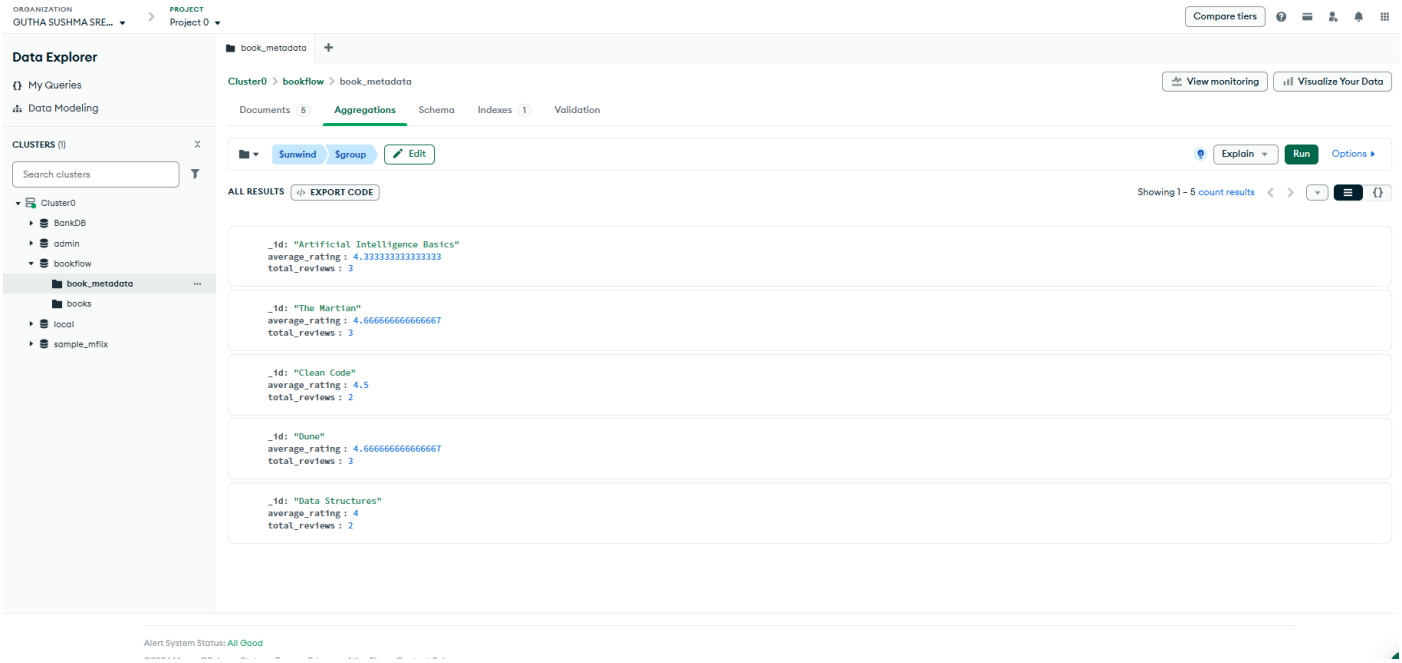
$unwind


ORGANIZATION
GUTHA SUSHMA SRE...
PROJECT
Project 0
Compare tiers
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
BankDB
admin
bookflow
book_metadata
books
looal
sample_mflix
book_metadata
Cluster0 bookflow book_metadata
View monitoring
Visualize Your Dat
Documents 5
Aggregations
Schema
Indexes 1
Validation
$unwind
Edit
Explain
Run
Options
ALL RESULTS
EXPORT CODE
Showing 1 - 13 count results
_id: ObjectId('6a7eb72ccd16c4c1e1e67084')
book_id : 101
title : "The Martian"
author : "Andy Weir"
published_year : 2021
category : "Science Fiction"
format : "Digital PDF"
file_size : "2.5 MB"
member_id : 101
reviews : Object (3)
_id: ObjectId('6a7eb72ccd16c4c1e1e67084')
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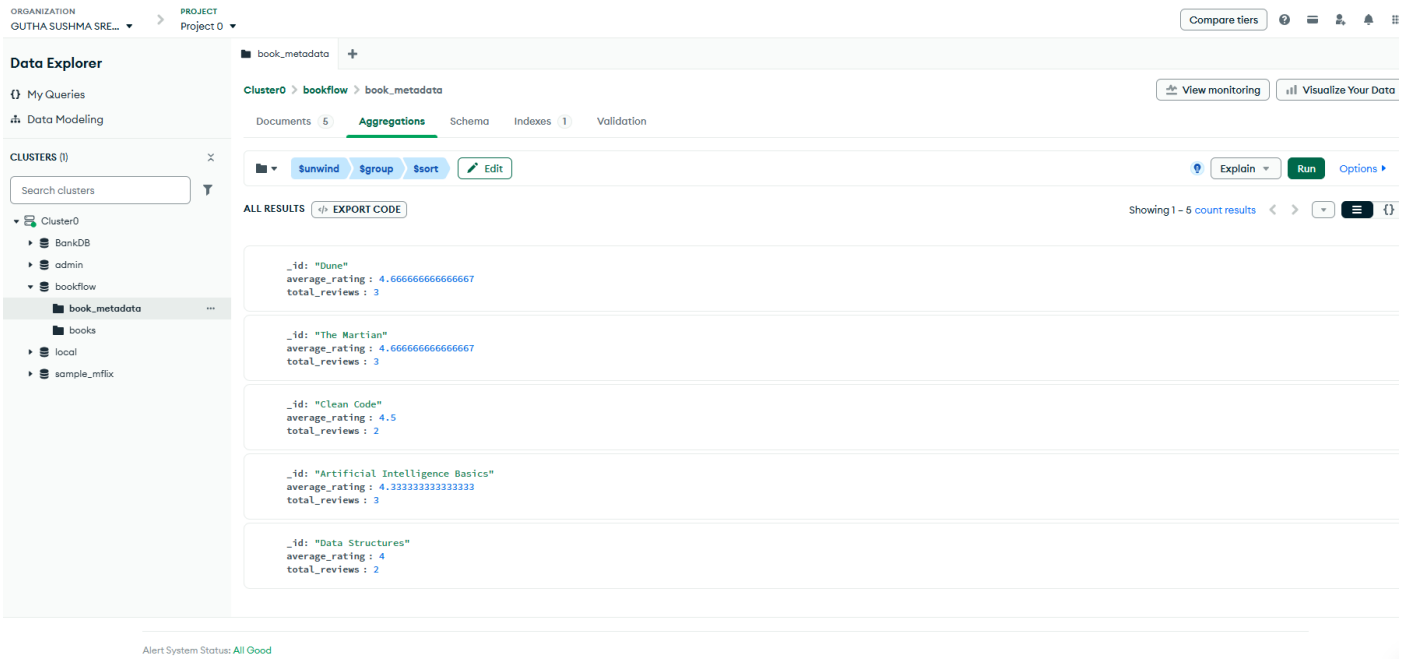

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Compare tiers
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My Queries
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CLUSTERS (1)
Search clusters
Cluster0
BankDB
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bookflow
book_metadata
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looal
sample_mflix
book_metadata
Cluster0 bookflow book_metadata
View monitoring
Visualize Your Data
Documents 5
Aggregations
Schema
Indexes 1
Validation
$unwind
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category : "Science Fiction"
format : "Digital PDF"
file_size : "2.5 MB"
member_id : 101
reviews : Object (3)
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author : "Frank Herbert"
published_year : 2022
category : "Science Fiction"
format : "Digital EPUB"
file_size : "3.1 MB"
member_id : 102
reviews : Object (3)
Alert System Status: All Good
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ORGANIZATION
GUTHA SUSHMA SRE...
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Compare tiers
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sample_mflix
book_metadata
Cluster0 > bookflow > book_metadata
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Visualize Your Data
Documents 5
Aggregations
Schema
Indexes 1
Validation
$unwind
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Showing 1 - 13 count results
_id: ObjectId('6a7eb759cd16c4c1e1e67087')
book_id : 102
title : "Dune"
author : "Frank Herbert"
published_year : 2022
category : "Science Fiction"
format : "Digital EPUB"
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member_id : 102
reviews : Object (3)
_id: ObjectId('6a7eb759cd16c4c1e1e67087')
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author : "Frank Herbert"
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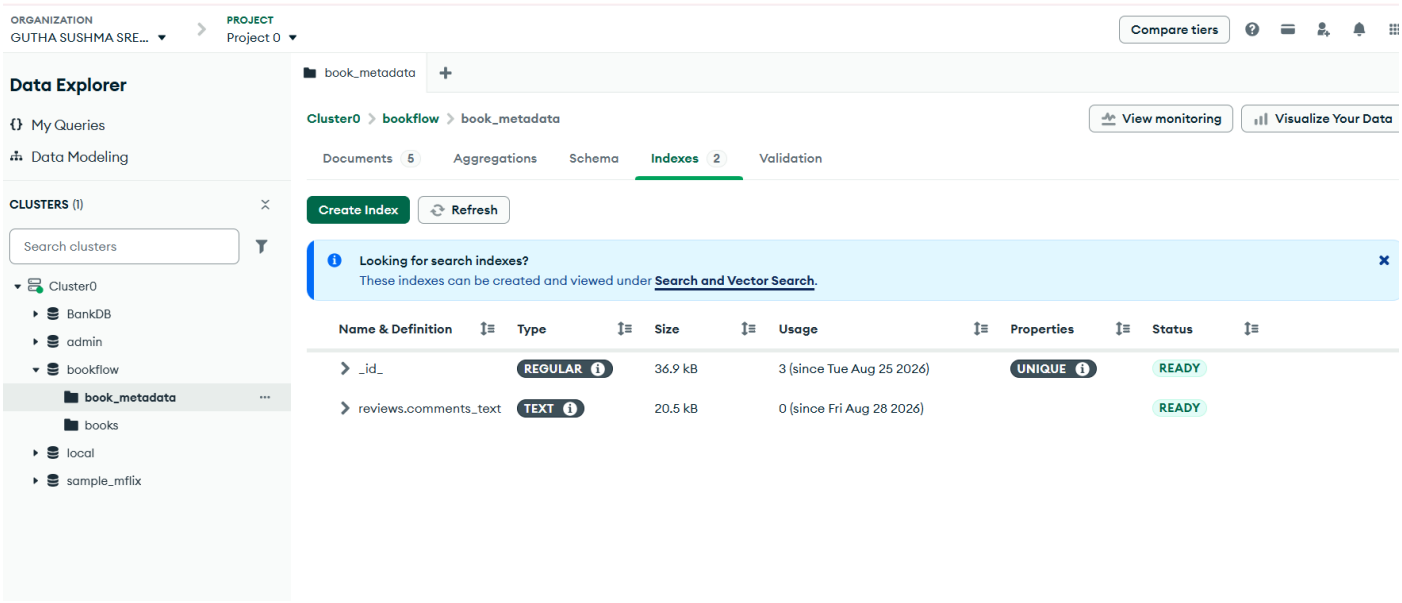
$group


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Cluster0 > bookflow > book_metadata
View monitoring
Visualize Your Data
Documents 5
Aggregations
Schema
Indexes 1
Validation
$unwind
$group
Edit
Explain
Run
Options
ALL RESULTS
EXPORT CODE
Showing 1 - 5 count results
_id: "Artificial Intelligence Basics"
average_rating : 4.333333333333333
total_reviews : 3
_id: "The Martian"
average_rating : 4.666666666666667
total_reviews : 3
_id: "Clean Code"
average_rating : 4.5
total_reviews : 2
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average_rating : 4
total_reviews : 2
Alert System Status: All Good

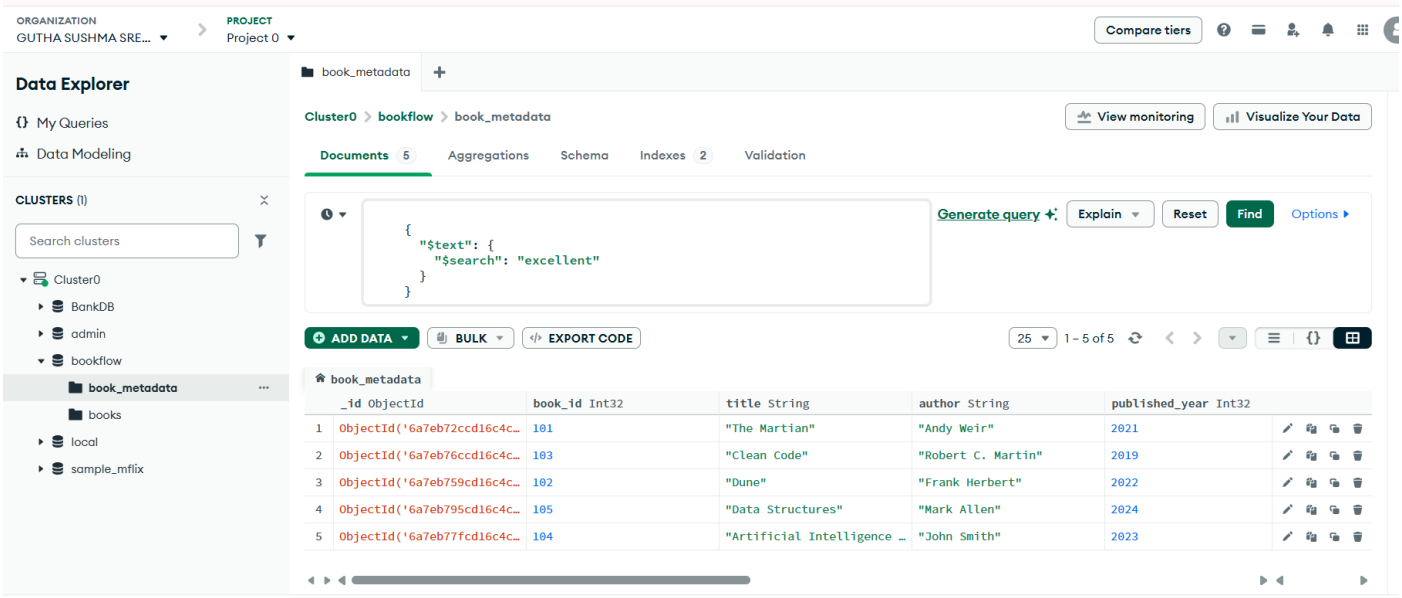
$sort



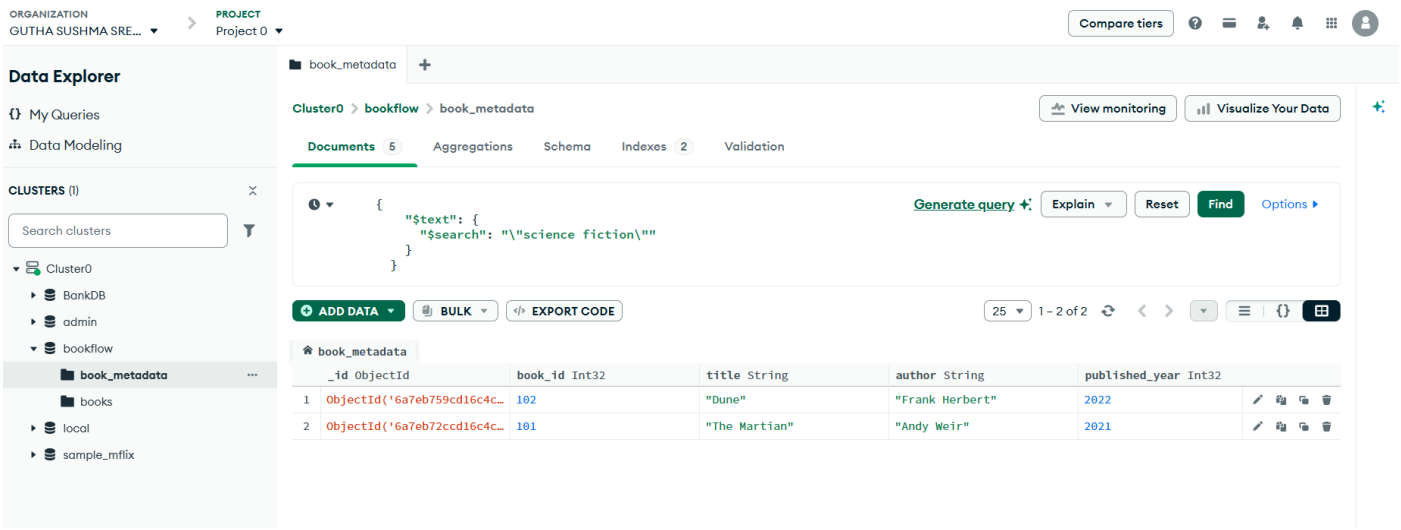
Create text index



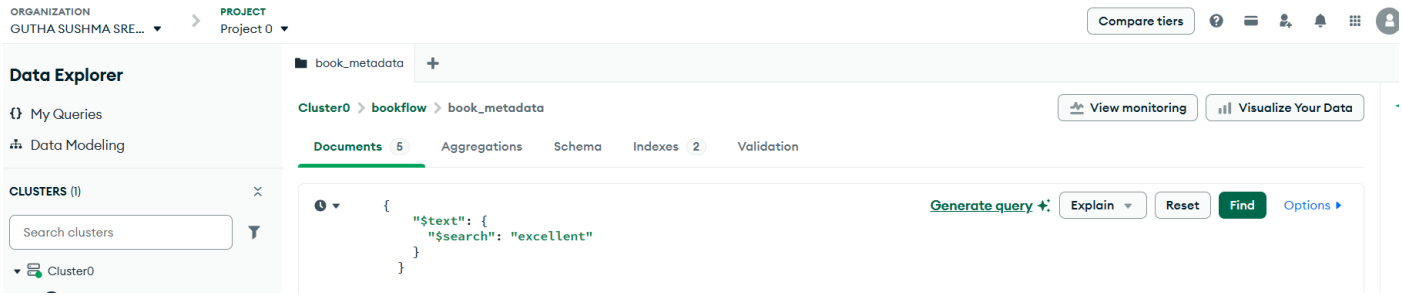
Search using text index



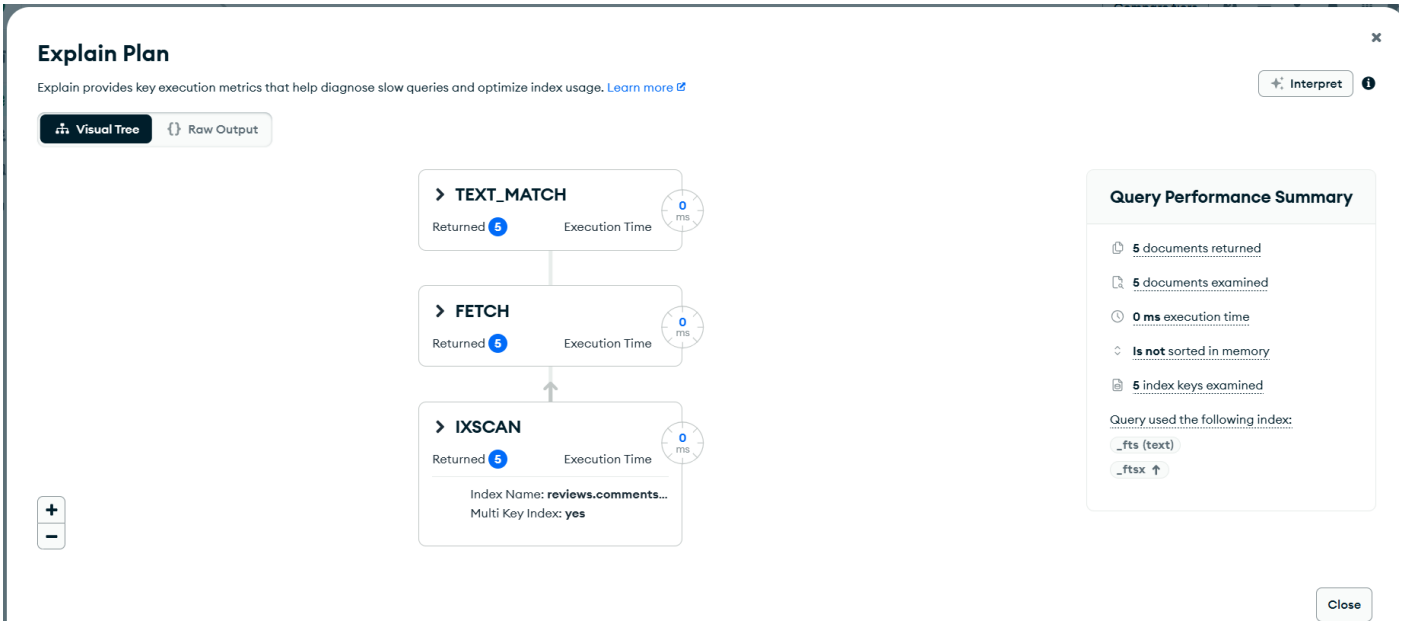
Search for specific keyword



Use explain()



Output:


Explain Plan
Explain provides key execution metrics that help diagnose slow queries and optimize index usage. Learn more
Interpret
Visual Tree
Raw Output
TEXT_MATCH
Returned 5
Execution Time
0 ms
FETCH
Returned 5
Execution Time
0 ms
IXSCAN
Returned 5
Execution Time
0 ms
Index Name: reviews.comments...
Multi Key Index: yes
Query Performance Summary
5 documents returned
5 documents examined
0 ms execution time
Is not sorted in memory
5 index keys examined
Query used the following index:
_fts (text)
_ftsx
Close